MedTrack_DV (Hospital Operations & Patient Analytics Dashboard)

Project Statement

This project aims to develop a comprehensive hospital operations and patient analytics dashboard suite for analyzing hospital performance, patient admissions, department efficiency, and healthcare resource utilization.

Using publicly available Hospital Datasets and Patient Admission Datasets, the project will transform healthcare operational data into actionable insights through interactive Tableau dashboards.

The dashboard suite will help hospital administrators, healthcare managers, medical staff, and policymakers monitor patient flow, evaluate department performance, optimize hospital resources, and improve operational efficiency through data-driven decision-making.

The final deliverable is a unified Tableau workbook consisting of four interconnected dashboards:

- Hospital Overview
- Patient Flow
- Department Analytics
- Resource Utilization

Outcomes

- Collection and integration of hospital datasets.
- Data cleaning and transformation for healthcare operations analytics.
- Development of hospital performance KPIs.
- Creation of four interactive Tableau dashboards.
- Patient flow and admission analysis.
- Department efficiency monitoring.

- Resource utilization tracking.
- Single Tableau workbook (.twbx) as final deliverable.

Modules to be Implemented

- Healthcare data collection
- Data cleaning and transformation
- KPI engineering
- Dashboard planning and wireframing
- Dashboard development
- Dashboard integration
- Testing and validation
- Documentation and project delivery

Workflow Diagram

```
Collect Hospital Datasets
            ↓
Data Cleaning & Transformation
            ↓
Healthcare KPI Engineering
            ↓
Dashboard Development
            ↓
Dashboard Integration
            ↓
Testing & Validation
            ↓
Documentation & Delivery
```

Milestone 1: Data Collection and Preparation (Weeks 1–2)

Module 1: Hospital Data Collection

Tasks

- Download hospital operational datasets.
- Collect patient admission records.
- Gather department and resource data.
- Integrate healthcare datasets.

Deliverables

- hospital_raw_data.csv
- data_collection.py

Evaluation

- Successful dataset integration.
- Dataset completeness above 95%.

Module 2: Data Cleaning & Transformation

Tasks

- Remove duplicate records.
- Handle missing patient data.
- Standardize department names.
- Normalize healthcare indicators.
- Create Tableau-ready datasets.

Deliverables

- hospital_cleaned.csv
- hospital_cleaning.ipynb

Evaluation

- Less than 2% missing values.

- Consistent operational metrics.

Milestone 2: KPI Engineering and Dashboard Planning (Weeks 3–4)

Module 3: Hospital KPI Engineering

Tasks

Calculate:

- Total Admissions
- Occupancy Rate
- Average Length of Stay
- Readmission Rate
- Bed Utilization Rate
- Department Efficiency Score

Deliverables

- hospital_final_dataset.xlsx
- generate_hospital_kpis.py

Evaluation

- All KPIs correctly calculated.
- Dataset optimized for Tableau.

Module 4: Dashboard Planning & Prototyping

Tasks

Design layouts for:

- Hospital Overview
- Patient Flow

- Department Analytics
- Resource Utilization

Define:

- Filters
- Navigation
- Dashboard actions
- Department comparisons

Deliverables

- dashboard_storyboard.pdf
- medtrack_prototype.twbx

Evaluation

- Dashboard designs approved.
- Prototype functionality verified.

Milestone 3: Dashboard Development (Weeks 5–6)

Module 5: Build Hospital Overview & Patient Flow Dashboards

Tasks

Hospital Overview

- Admissions overview
- Occupancy monitoring
- Readmission analysis
- Hospital performance KPIs
- Monthly operational trends

Patient Flow

- Admission trends
- Discharge tracking
- Patient movement analysis
- Average stay analysis
- Peak patient load monitoring

Deliverables

- medtrack_dashboard_v1.twbx

Evaluation

- Interactive visualizations operational.
- Hospital KPIs validated.

Module 6: Build Department Analytics & Resource Utilization Dashboards

Tasks

Department Analytics

- Department performance analysis
- Patient volume by department
- Readmission by department
- Department efficiency comparison
- Treatment capacity analysis

Resource Utilization

- Bed utilization analysis
- Staff allocation monitoring
- Equipment utilization tracking
- Capacity planning insights

- Resource availability analysis

Dashboard Integration

- Global filters
- Navigation controls
- Parameter actions
- Dashboard linking

Deliverables

- MedTrack_DV.twbx

Evaluation

- All dashboards integrated.
- Navigation and filters functioning correctly.

Milestone 4: Testing and Delivery (Weeks 7–8)

Module 7: Testing and Validation

Tasks

- Validate KPI calculations.
- Verify healthcare metrics.
- Test dashboard interactions.
- Validate patient flow analytics.

Deliverables

- QA Checklist
- Dashboard Testing Report

Evaluation

- No major dashboard issues.
- KPI accuracy above 95%.

Module 8: Documentation and Project Delivery

Tasks

Prepare project documentation including:

- Dataset sources
- KPI definitions
- Dashboard guide
- Healthcare operations methodology

Organize project structure:

```
/scripts
/data
/dashboard
/docs
```

Deploy:

- GitHub Repository
- Tableau Public (Optional)

Deliverables

- GitHub Repository
- Final Documentation
- Tableau Workbook

Evaluation

- Fully documented project.
- Portfolio-ready dashboard suite.

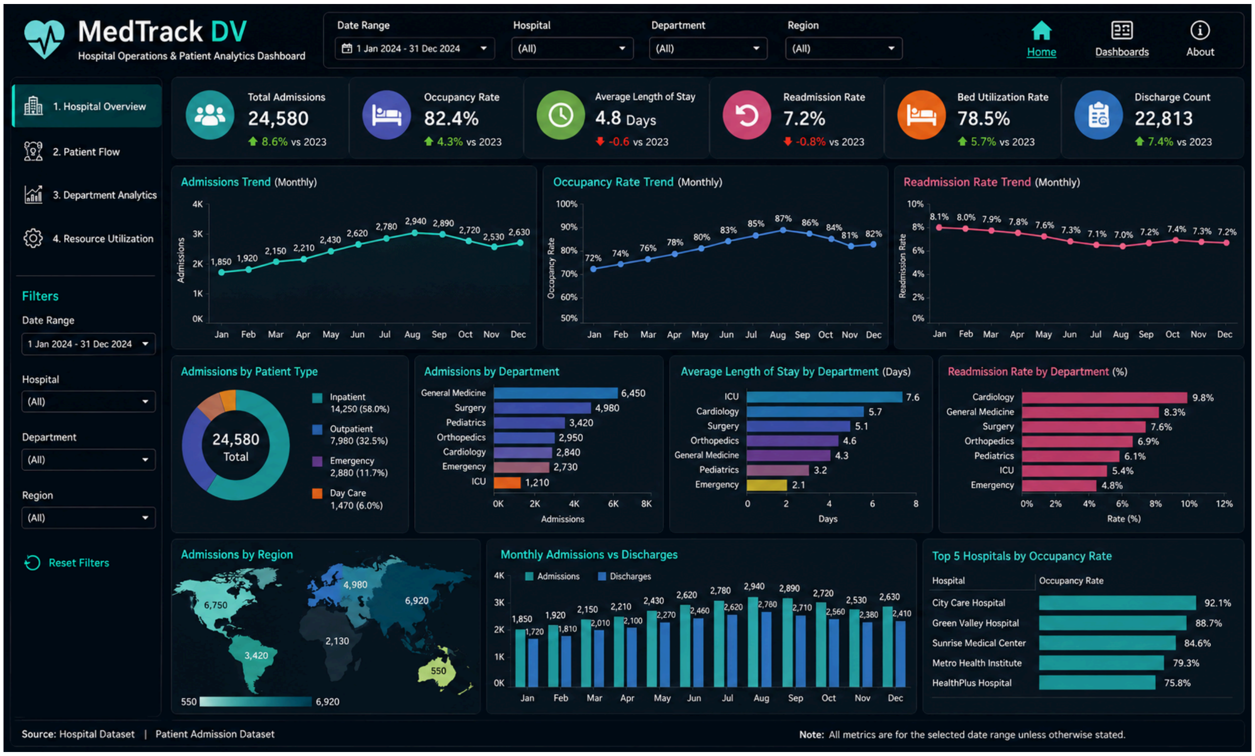
Evaluation Criteria

Milestone	Focus Area	Metric	Target
1	Data Collection & Cleaning	Dataset Completeness	>95% Complete
2	KPI Engineering	KPI Accuracy	6+ KPIs Generated
3	Dashboard Development	Dashboard Functionality	4 Dashboards Integrated
4	Documentation & Delivery	Project Quality	Portfolio Ready

Tech Stack

Area	Tools / Libraries
Data Collection	Python, Hospital Dataset, Patient Admission Dataset
Data Processing	Pandas, NumPy
Data Cleaning	Python
Visualization	Tableau Desktop / Tableau Public
Dashboard Integration	Tableau Filters, Parameters, Actions
Documentation	Markdown, GitHub

Sample Dashboard Layout:


MedTrack DV
Hospital Operations & Patient Analytics Dashboard
Date Range
1 Jan 2024 - 31 Dec 2024
Hospital
(All)
Department
(All)
Region
(All)
Home
Dashboards
About
1. Hospital Overview
2. Patient Flow
3. Department Analytics
4. Resource Utilization
Filters
Date Range
1 Jan 2024 - 31 Dec 2024
Hospital
(All)
Department
(All)
Region
(All)
Reset Filters
Total Admissions
24,580
8.6% vs 2023
Occupancy Rate
82.4%
4.3% vs 2023
Average Length of Stay
4.8 Days
-0.6 vs 2023
Readmission Rate
7.2%
-0.8% vs 2023
Bed Utilization Rate
78.5%
5.7% vs 2023
Discharge Count
22,813
7.4% vs 2023
Admissions Trend (Monthly)
Occupancy Rate Trend (Monthly)
Readmission Rate Trend (Monthly)
Admissions by Patient Type
24,580 Total
Inpatient 14,250 (58.0%)
Outpatient 7,980 (32.5%)
Emergency 2,880 (11.7%)
Day Care 1,470 (6.0%)
Admissions by Department
General Medicine 6,450
Surgery 4,980
Pediatrics 3,420
Orthopedics 2,950
Cardiology 2,840
Emergency 2,730
ICU 1,210
Average Length of Stay by Department (Days)
ICU 7.6
Cardiology 5.7
Surgery 5.1
Orthopedics 4.6
General Medicine 4.3
Pediatrics 3.2
Emergency 2.1
Readmission Rate by Department (%)
Cardiology 9.8%
General Medicine 8.3%
Surgery 7.6%
Orthopedics 6.9%
Pediatrics 6.1%
ICU 5.4%
Emergency 4.8%
Admissions by Region
Monthly Admissions vs Discharges
Admissions
Discharges
Top 5 Hospitals by Occupancy Rate
Hospital
Occupancy Rate
City Care Hospital 92.1%
Green Valley Hospital 88.7%
Sunrise Medical Center 84.6%
Metro Health Institute 79.3%
HealthPlus Hospital 75.8%
Source: Hospital Dataset | Patient Admission Dataset
Note: All metrics are for the selected date range unless otherwise stated.